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Hierarchical DFT in a RISC-V processor

Executive summary

This paper demonstrates the application of the Tessent™ hierarchical DFT and ATPG methodology on a RISC-V processor. With hierarchical DFT, all the DFT is completed at the block level and then replicated to the top level. Using a RISC-V test case, we show how to insert Logic BIST, Memory BIST, scan cells, TAP, and an in-system controller as well as perform ATPG. Hierarchical DFT moves the DFT efforts early in the design flow and reduces ATPG runtime and compute resources by 10X.

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Introduction

The RISC-V (Reduced Instruction Set Computer) is the only processor with a completely open source instruction set, and is supported by a broad industry ecosystem of at least 70 companies. The use of RISC-V is growing in popularity in many market sectors because it lends flexibility, scalability, and extensibility to SoCs, and eliminates hardware license fees.

RISC-V is increasingly being used in IC designs for artificial intelligence (AI), embedded, graphics engines, networking, storage, and security. Design tools, like the Tessent DFT family, are in place to support designs that include RISC-V cores. Because these processors are used as cores in large, complex designs, it's important to implement strategies that reduce DFT and ATPG time and cost.

This paper demonstrates the application of the Tessent hierarchical DFT and ATPG methodology on a RISC-V processor. With hierarchical DFT, all the DFT is completed at the block level and then replicated to the top level. Using a RISC-V test case, we show how to insert Logic BIST, Memory BIST, scan cells, TAP, and an in-system controller as well as perform ATPG. Hierarchical DFT moves the DFT efforts early in the design flow and reduces ATPG runtime and compute resources by 10X.

We focus on how to implement a hierarchical DFT methodology so the test logic insertion and test pattern generation can be completed at the RISC-V core level and then retargeted to the top level. Some designs do not allow for new levels of hierarchy or top-level logic. Therefore, we also show how to embed DFT logic for block interconnect test and TAP within one of the existing blocks. This example shows this methodology with very high coverage results of >99% stuck coverage and >93% transition coverage.

Note that the RISC-V processor is likely embedded with additional logic. The example design we use here treats the processor as if it is the top-level design. If it were embedded in a bigger design, then the same hierarchical pattern retargeting would be applied from the core but to the design top level.

The RISC-V test case

We downloaded small design from <https://github.com/ataradov/riscv/tree/master/rtl> to use for this hierarchical DFT demonstration. Tessent customers can get an example of this testcase from their application engineer. It shows automation for all the steps described in this paper. Figure 1 illustrates a block diagram of the design.

The top SoC consists of a RISC-V core, a memory block, and four more blocks. There is one clock, one reset and 66 other input/outputs. The design has 1,769 flops in total. The RTL copyright information is given with the distribution of the RTL code.

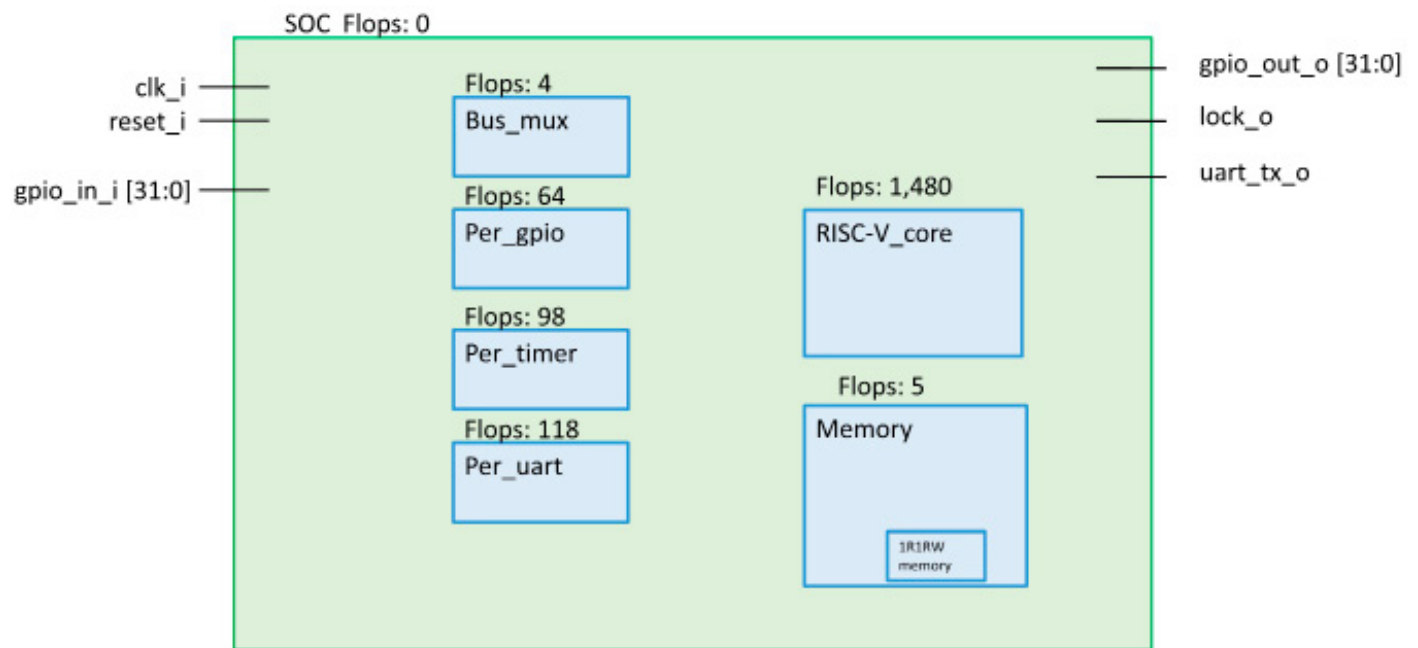


Figure 1: Design block diagram.

DFT strategy

This section describes the actions taken to implement hierarchical DFT on the RISC-V core, the memory, and for the top-level SoC. The Tessent DFT products used in this example are (software version: v2019.2):

- Tessent ScanPro: OCC, TPI, X-bounding, and scan insertion
- Tessent TestKompress™: ATPG compression
- Tessent LogicBIST: Logic BIST
- Tessent MemoryBIST: Memory BIST
- Tessent BoundaryScan: Boundary scan
- Tessent MissionMode: In-system controller

Hierarchical DFT and ATPG for the RISC-V core

We applied hierarchical test, starting with the addition of a wrapper chain and OCC to enable hierarchical test. Next is the addition of ATPG compression scan and Logic BIST. Patterns generated by ATPG (INTEST) and Logic BIST will be applied from the SoC level to the RISC-V through a process called pattern retargeting. All DFT IPs are added within existing blocks.

Memory BIST

We added memory BIST to the memory core, which is accessed from the SoC through the test access mechanism (TAP) or the Tessent MissionMode controller.

SoC top level DFT And pattern retargeting

The generated ATPG and logic BIST patterns for the RISC-V core test are retargeted to the SoC level. Another ATPG is run for the other blocks and the RISC-V's wrapper chain together (EXTEST). We then add an IEEE 1149.1 TAP to control boundary scan, ATPG, logic BIST, and memory BIST. For in-system test, the Tessent MissionMode controller takes over the TAP to access logic BIST and memory BIST. All DFT IPs are added within the "Per_uart" block.

Test case structure and tasks

We organized the testcase to make it easy to understand the library and file structure as well as the execution steps. The directory structure is shown on the next page. An application engineer can provide the testcase.

```

/ Lib
/ dft_sim.atpg: Tessent cell library
/ memory_core.tcd_mem_lib: MemoryBIST library
/ dft_sim.v: Verilog library
/ memory_core.v: Memory Verilog library
/ rom.v: MissionMode ROM Verilog library
/ power_on_reset_generator.v: Power on reset Verilog library / RTL
/ bus_mux.v: Verilog RTL for "bus_mux" block
/ memory.v : Verilog RTL for "memory" block
/ per_gpio.v : Verilog RTL for "per_gpio" block
/ per_timer.v : Verilog RTL for "per_timer" block
/ per_uart_.v : Verilog RTL for "per_uart" block
/ riscv_core.v : Verilog RTL for "riscv_core" block
/ soc_pad.v : Verilog RTL for "soc" top
/ Work
/ RISCv_core_dir: Working directory for RISCv_core task
/ 01_TKLBIOT_OCC: EDT and Logic BIST generation and insertion
/ 02_TS_synthesis: Logic synthesis to riscv_core.v and the inserted DFT IPs (converting RTL
to a gate design)
/ 03_TPI: Test Point Insertion
/ 04_SCAN_X_BOUNDING: X-bounding and Scan Insertion
/ 05_LBIOT_FSIM: LBIOT Fault Simulation
/ 06_LBIOT_SIM: Pattern verification
/ 07_LBIOT_RET_and_SIM: LBIOT pattern retargeting and verification
/ 08_TK: TestKompress ATPG (INTEST)
/ 09_TK_SIM: Pattern verification
/ Tsdb_dir: Tessent Database
/ SoC_dir: : Working directory for memory and SoC task
/ 01_BS_TAP_MBIST_MM: Boundary Scan, TAP, MBIST, MissionMode generation and insertion
/ 02_TK_OCC: EDT and OCC insertion and generation
/ 03_TS_Synthesis: Logic synthesis to soc and the inserted DFT IPs
/ 04_SCAN: Scan Insertion
/ 05_LBIOT_RET_and_SIM: LBIOT pattern retargeting and verification
/ 06_TK: TestKompress ATPG (EXTEST)
/ 07_TK_SIM: Pattern verification
/ 08_RISCv_core_ATPG_RET: RISCv_core ATPG (INTEST) pattern retargeting
/ 09_RISCv_core_ATPG_RET_SIM: Pattern verification
/ 10_BS_MBIST_RET_and_SIM: MBIST pattern retargeting and verification + pattern verification
for Boundary scan
/ 11_MM_RET_and_SIM: MissionMode (LBIOT and MBIST) pattern retargeting and verification

```


Final DFT-inserted design

Figure 2 shows the RISC-V design with all Tessent DFT IPs in place. We added BIST to the 1R1WR memory under the Memory block, and added scan compression, logic BIST, and wrapper chain to the RISC-V_core block. The memory BIST and logic BIST are accessed from either the TAP or the MissionMode controller. On the SoC, embedded deterministic test (EDT) was added under the "Per_uart" block to test all the logic except

the logic already tested by the scan/logic BIST under the RISC-V_core (INTEST). Two OCCs were added; one is in the RISC-V_core for scan pattern retargeting (INTEST) and the other is in "Per_uart" block (EXTEST). The OCC under "Per_uart" is bypassed; the OCC under RISC-V is used during INTEST. Low power shift (toggle range $\leq 15\%$) is used for both ATPG and logic BIST.

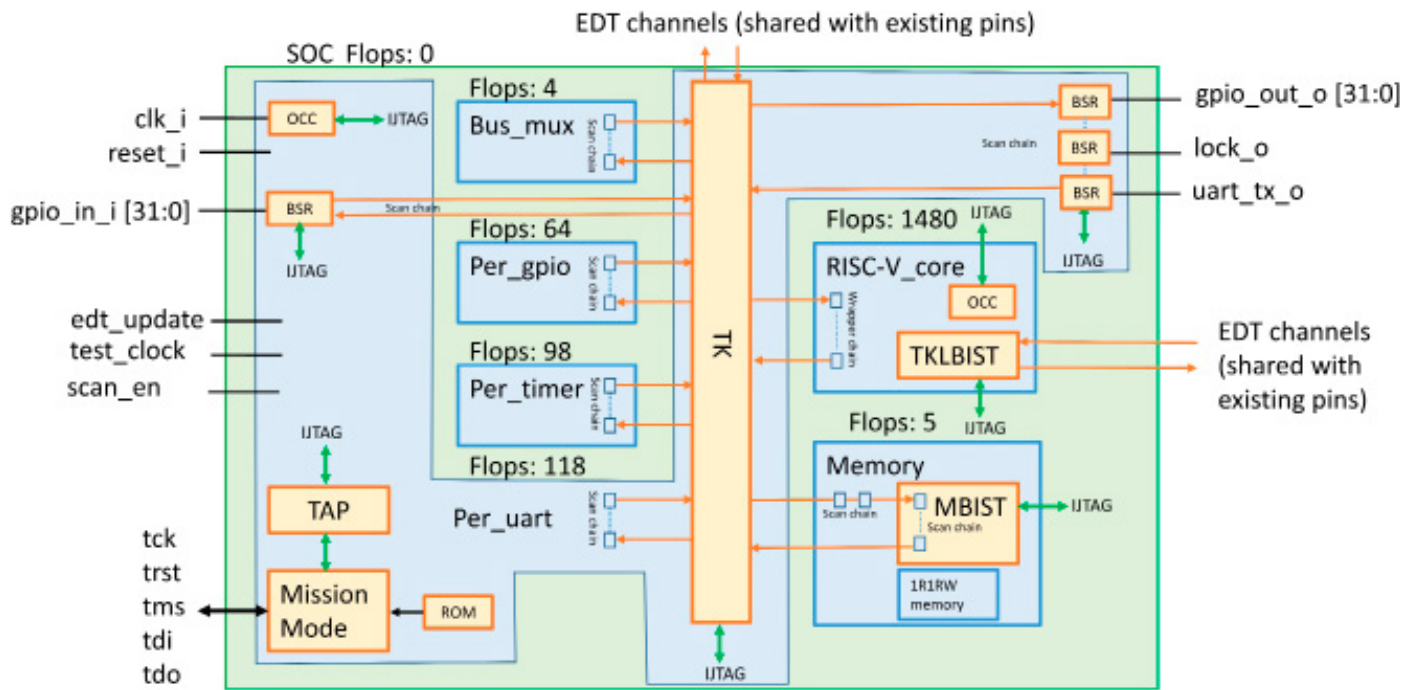


Figure 2: Design block diagram after all DFT logic is added.

Results

We collected results of the scan/EDT configuration and coverage levels for this test case.

Table 1 – Scan/EDT configuration.

	Scan chain	Scan length	Scan cell	EDT Ins	EDT Outs	Init cycles	Low power shift
RISC-V_core (ATPG or LBIST)	20	97	1,925	3	3	22	15%
15%	22	50	819	3	3	17	15%

Table 2 shows the stuck-at and transition test coverage for this design.

Table 2 – The stuck-at and transition pattern generation reach very high coverage.

	Stuck-at			Transition		
	Test Coverage	Pattern count	CPU(s)	Test Coverage	Pattern Count	CPU(s)
ATPG Intest	97.44%	253	20	95.17%	488	35
ATPG total	99.10%	599	95	93.80%	1,482	213
Logic BIST	93.45	1,000	13	85.64%	3,000	14

Tessent MissionMode Controller

Table 3 shows how many test cycles were required to run logic BIST and memory BIST through the Tessent MissionMode controller instead of the TAP. It also shows how much memory was needed to store the IJTAG instructions used by the MissionMode controller.

Table 3 – Test cycles and memory requirements for testing through the MissionMode controller.

	Stuck-at			Transition		
	Test Coverage	Pattern count	CPU(s)	Test Coverage	Pattern Count	CPU(s)
Logic BIST	99.10%	599	95	93.80%	1,482	213
Logic BIST	93.45	1,000	13	85.64%	3,000	14

Conclusion

As RISC-V processors come into greater use, DFT designers will need test tools and flows that support RISC-V processors in context of large, complex SoCs. This paper demonstrates how to apply the Tessent hierarchical DFT to a RISC-V processor. Designers who are using RISC-V processors and Siemens EDA DFT tools can access the full test case including scripts from their Tessent application support representative.

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